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Seed priming with chitosan nanoparticles improved seed vigour parameters and antioxidant enzyme kinetics under polyethylene glycol (PEG) induced drought stress in cotton

V Santhy¹, Pratik Dhakade¹, Debashis Paul^{1*} , Rakesh Kumar¹, Ajit Meshram¹, Sunil Mahajan¹ and V. N. Waghmare¹

Abstract

Global cotton production often experiences significant obstacles from climatic variability, particularly due to drought stress. The present study evaluated the efficacy of seed priming with chitosan nanoparticles (CNPs) as a sustainable strategy to mitigate drought stress in cotton under rainfed ecology of cotton cultivation in India. The laboratory synthesized chitosan nanoparticles using the ionic gelation method with trisodium polyphosphate (TTP) as a crosslinker were characterized through transmission electron microscopy (Joel JEM 2100 Plus) and particle size analyser (Malvern Zetasizer Nano ZS90). *Bt*-cotton (*Bacillus thuringiensis* derived toxin; GMO cotton) seeds were then subjected to hydropriming and 200 ppm chitosan nanoparticles (CNPs) priming along with an untreated control under laboratory conditions at ambient temperature (27 °C) with three independent runs. Seedling growth parameters, seed vigour indices, germination (%) and key biochemical markers such as total protein content, catalase, superoxide dismutase and peroxidase enzyme activity were recorded under varying PEG (polyethylene glycol) induced stress conditions. Our findings highlighted that CNPs priming significantly enhanced all vigour parameters under optimal (no stress) conditions. Chitosan nanoparticle (CNPs) primed seedlings maintained significantly higher root length (10.80 cm), shoot length (11.27 cm), dry weight (823.33 mg), and germination percentage (96.33%) under 5% PEG stress, compared to hydro primed and untreated seeds. This enhanced performance may be attributed to significantly higher total protein content and higher catalase and peroxidase activity in CNPs primed seeds. The positive effects of CNPs were further validated in the soil tube experiment with moisture level of 100%, 75% and 60% field capacity, demonstrating enhanced root and shoot development with chitosan priming under lowered Field capacity compared to untreated seeds. Seed priming with CNPs offers a sustainable strategy for drought tolerance in cotton, promoting seed vigour traits and pre-activating biochemical responses. This chitosan nanoparticle seed priming approach is highly promising for enhancing crop resilience in water scarce environments.

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Keywords Chitosan nanoparticles (CNPs), Water scarcity, Seed priming, Crop resilience, Cotton

Introduction

Cotton (*Gossypium hirsutum* L.) is a vital global commodity, providing the world's leading natural fibre and contributing significantly to the oil industry. It serves as a major contributor to the economies of many developing countries supporting millions of rural livelihoods. The world cotton production for the year 2025 is projected as 25.86 million tonnes (MT) contributing mainly by China (6.3 MT), India (5.0 MT) and USA (3.09 MT) (World: ICAC Journal 'Cotton This Month' – 1st July 2025). Climate change is one of the major challenges in cotton production and, low lint productivity per unit area in some regions such as India (448 kg/ha.), Indonesia (352 Kg/ha.) and few African countries (less than 300 Kg/ha) (ICAC Cotton data book 2025; <https://icac.org/wp-content/uploads/2023/06/DATABOOK-2025-sample.pdf>) is affecting global cotton output. The cotton cultivation is highly vulnerable due to increasing unpredictability of precipitation patterns caused by climate change severely constraining growth, yield and fibre quality. It is imperative to have effective strategies for better resilience in cotton so as to ensure global food and economic security. Among the abiotic stresses drought is considered the most detrimental adversely affecting total crop productivity. Moisture stress during critical stages of germination and seedling establishment impairs water uptake, disrupts metabolic functions, and induces oxidative damage, leading to poor crop stand and ultimately significant economic losses [1]. Previous researchers [2] reported drought affects various physiological processes, such as rate of photosynthesis, stomatal regulation and reduction in root-shoot growth, reduced leaf area expansion, reduced transpiration and osmoregulation in cotton which result into square and boll shed and low lint yield. Earlier report [3] also documented severe reduction in seedling length and photosynthetic rate in cotton. Seed quality improvement through priming is an important area of research in crop stress management for enhancing seedling vigour and conferring drought tolerance [4].

Polyethylene glycol (PEG-6000) is a widely used osmotic medium for simulating and evaluation of drought stress in crops because its high molecular weight prevents it from penetrating plant cell wall [5, 58]. In crops such as, wheat, rice and buckwheat, PEG-induced stress significantly reduces seed germination percentage and seedling biomass [6]. Earlier findings indicate that increasing PEG concentrations leads to excessive production of reactive oxygen species (ROS), causing oxidative damage and reduced chlorophyll levels, which ultimately impairs photosynthetic efficiency [7]. Earlier researchers [8] also observed PEG negatively impacts shoot and root

length in legumes like green gram and chickpea. Earlier studies also used PEG for inducing drought stress to screen tolerant genotypes in cotton [9], in cowpea [10], in green gram [11] and in sugarcane [12].

Seed priming, a pre-sowing hydration technique, is a simple, cost-effective, and eco-friendly approach to improve germination and stress tolerance in numerous crops [13, 14]. Techniques like hydropriming (soaking in water) can activate pre-germinative metabolic processes, preparing the seed for a more rapid and synchronized emergence [15]. However, the efficacy of conventional priming can be limited and inconsistent. Recent advances in nanotechnology offer a novel frontier for improving priming protocols. Nanoparticles, due to their small size, high surface area-to-volume ratio, and unique physicochemical properties, can interact with biological systems in novel ways, enhancing nutrient uptake and triggering robust defence mechanisms [16]. The potential of nanoparticles in promotion of seed vigour through seed priming has been reported in cucumber [17], wheat [18], maize [19], rice [20] and garden pea [21] under various abiotic stress conditions.

In cotton, seed priming with zinc oxide nanoparticles showed efficacy in enhancing Crop tolerance to environmental stresses [22]. Earlier findings [23] also noted nano priming with carbon dots enhances cotton seed germination and salt tolerance by activating salt-induced ROS signalling to modulate Na⁺ homeostasis. Previous studies [24] also noted seed priming with cerium oxide nanoparticles enhances the plant resilience to salinity stress.

Among various biopolymers, chitosan, a deacetylated derivative of chitin, has gained considerable attention for its excellent biodegradability, biocompatibility, and potent eliciting activity [62]. Chitosan nanoparticles (CNPs) combine the intrinsic bioactivity of chitosan with the advantages of nano-dimensions, facilitating better absorption by seeds and more efficient interaction with cellular components [25]. Previous studies have suggested that CNPs seed priming can be a sustainable approach towards plant stress mitigation in various crops [26–28]. In India majorly cotton growing states like Gujarat, Maharashtra, Madhya Pradesh (covering approximately 7.5-million-hectare area out of 12 million hectare) are prone to drought stress. Therefore, seed priming with potential nanoparticles may be a sustainable way to minimize adverse drought effect. However, in cotton, potential of chitosan nanoparticles as a seed priming agent to mitigate drought stress during the critical seedling establishment phase remains underexplored.

Therefore, this study was designed to evaluate and compare the effectiveness of seed priming with chitosan

nanoparticles (CNPs) and hydropriming in enhancing drought tolerance in cotton. We hypothesized that CNPs priming would superiorly ameliorate the negative effects of PEG-induced osmotic stress by improving early seedling attributes, enhancing seed vigour, and modulating key biochemical defence responses. The specific objectives were to investigate the effects of these priming treatments on (1) seedling growth traits (root and shoot length, dry weight), (2) seed vigour indices and germination potential, and (3) biochemical markers (total protein content and peroxidase activity) under PEG induced drought stress conditions.

Materials and methods

Plant materials and chemicals

The seeds of cotton genotype (Suraj *Bt*) were procured from the Seed Technology section, ICAR-Central Institute for Cotton Research, Nagpur, Maharashtra. Chitosan, TTP (Tri-sodium polyphosphate), and Sodium hydroxide pellets were purchased from Sigma to synthesize Chitosan Nanoparticles.

The freshly harvested seeds of Suraj *Bt* were first ginned (the process of removal of fibre/lint and seed from the seed cotton), then the fuzzy seeds were processed through acid delinting [31]. Then the delinted seeds were undergone seed processing through gravity separator to increase the pure seed percentage. After that the seeds were dried uniformly and used for priming treatment. This total process takes around one month time from harvesting to seed priming treatments.

Preparation of CNPs

Chitosan nanoparticles (CNPs) were synthesized using the ionic gelation method with trisodium polyphosphate (TTP) as a crosslinker. A chitosan solution was prepared at a concentration of 0.5% (w/v) by dissolving chitosan in a 0.5% (v/v) aqueous acetic acid solution. This mixture was stirred overnight at 250 rpm on a magnetic stirrer at 25 °C. To prepare the TTP crosslinker, 0.25% (w/v) of TTP was dissolved in deionized water and then filtered through a 0.22 µm syringe filter. The TTP solution was added dropwise to the chitosan solution while stirring at 700 rpm at 25 °C for approximately 60 min to create the CNPs. The pH of the solution was maintained at 5.4 ± 0.2 throughout the synthesis process. After synthesis, the nanoparticles were centrifuged at 11,337 g for 20 min at 4 °C [29]. The pelleted CNPs were then resuspended in an aqueous acetic acid solution adjusted to pH 5.4 ± 0.2 and kept it at 4 °C for further use.

Pellets were used for further physicochemical characterization, which involves Dynamic Light Scattering (DLS), Zeta Potential (Malvern Zetasizer Nano ZS90), Transmission Electron Microscope (TEM; Joel JEM 2100 Plus), FTIR (Fourier Transform Infrared Spectroscopy;

Diamond alpha ATR, Bruker optics, Germany GmbH) and UV spectroscopy (Shimadzu uv- 1900i).

Preparation for osmotic stress condition

Osmotic stress was applied to cotton seeds by preparing aqueous solutions of polyethylene glycol (PEG) 6000 at concentrations of 1% (w/v) and 5% (w/v) as also screened for drought tolerance studies in cotton by earlier researchers [56–58]. Germination paper was then imbibed for 12 h with these PEG solutions to induce the stress conditions prior to the germination of both untreated and treated cotton seeds.

Priming treatments

The seeds of Suraj *Bt* there after undergone hydropriming and 200 ppm CNPs priming treatments for 2 h to initiate the early stages of imbibition along with untreated control as per the method demonstrated [30]. Seeds were subjected to various chitosan nanoparticle treatment concentrations (100–3000 ppm); based on the preliminary in-house results (unpublished; representative images provided as supplementary Fig. 1 [59]), , 200 ppm CNPs priming treatment was adopted for the experimental trials as we found 200 ppm concentration of chitosan nano particle was superior over other concentrations.

Seed quality parameters

Germination percentage (as per ISTA 2019)

One hundred seeds (100) of every treatment in three replications were placed in between sufficient moistened rolled towel papers and kept at 25 °C in seed germinator (iGene LabServe) with 12-hour light and 12-hour dark conditions. Final count was recorded on 12th day. The number of normal seedlings (with optimum root and shoot development) were at final count was expressed as percentage.

Root length (cm), Shoot length (cm), Seedling dry weight (mg)

Root length (cm) and Shoot length (cm) of ten randomly selected normal seedlings from towel paper in three replication was recorded and average root and shoot length was expressed in centimetres (cm). Fresh weight was calculated by taking ten normal seedlings whose length was measured and expressed in grams. Ten normal seedlings whose fresh weight was recorded were dried in a hot air oven to a constant weight at 80 ± 1 °C for three days to calculate the dry weight (mg).

Seed index was estimated as weight of one hundred (100) delinted seeds and expressed in gram (g) as described by Paul et al. [31].

Seed vigour index I and Seed vigour index II were calculated by the method suggested by Abdul-Baki and Anderson [32].

Seed Vigour Index–I= Germination (%) × average seedling length (cm).

Seed Vigour Index–II= Germination (%) × average seedling dry weight (mg).

Biochemical parameters

Total protein (mg/g)

Total protein concentration of the chitosan nanoparticles was determined using the Bradford assay [33] with a standard curve generated from bovine serum albumin (BSA). Samples were homogenized in 1 mL of 50 mM Tris-HCl (pH 7.5) with 1 mM EDTA and 1% PVP and centrifuged ($12,000 \times g$, 15 min, 4 °C), and 0.1 mL supernatant was mixed with 5 mL Bradford reagent. Absorbance was recorded at 595 nm, and protein content was expressed as mg g⁻¹ FW (Bovine Serum Albumin (BSA) equivalents).

Peroxidase activity (unit activity ml⁻¹ min⁻¹; POX)

Peroxidase activity was measured spectrophotometrically by monitoring the increase in absorbance at 470 nm for 2 min, which corresponds to the oxidation of guaiacol. The reaction mixture (3.12 mL total volume) consisted of 2.47 mL of 50 mM sodium phosphate buffer (pH7.8), 100 µL of enzyme extract, and 500 µL of 30 mM guaiacol. The reaction was initiated by adding 5 µL of a 30% hydrogen peroxide (H₂O₂) solution [34, 20]. One unit of peroxidase activity was defined as the amount of enzyme that caused an increase in absorbance of 0.01 per minute under the specified assay conditions.

Superoxide dismutase (unit activity ml⁻¹; SOD)

Superoxide dismutase (SOD) activity was determined by its ability to inhibit the photoreduction of nitro blue tetrazolium (NBT) [35]. The 3 mL reaction mixture was composed of 50 mM sodium phosphate buffer (pH7.8), 5 mM EDTA (pH7.5), 130 mM methionine, 7.5 mM NBT, and 100 µL of enzyme extract. The reaction was initiated by adding 2 mM riboflavin [36] and was incubated for 15 min under light. The absorbance of the reaction mixture was then measured at a wavelength of 560 nm. One unit of SOD activity was defined as the amount of enzyme that caused a 50% inhibition of the photoreduction of NBT under these assay conditions.

Catalase (unit activity ml⁻¹ protein min⁻¹)

The conversion of hydrogen peroxide to water and molecular oxygen is catalyzed by catalase. Because of its greater K_m value for H₂O₂ than ascorbate peroxidase (APX), it is thought to be a less effective mechanism for scavenging H₂O₂. The reaction mixture consisted of 1.5 mL of 100 mM potassium phosphate buffer (pH 7.0) to obtain a final concentration of 50 mM, 0.5 mL of 75 mM hydrogen peroxide to achieve a final concentration

of 12.5 mM, and 50 µL of enzyme solution. The catalase assay relies on the absorbance of H₂O₂ in the UV spectrum at 240 nm. According to previous researcher [37], a decline in absorbance is observed over time. The concentration of hydrogen peroxide reduced (first reading - final reading = amount of H₂O₂ reduced) per minute, per milligram of protein is used to calculate enzyme activity.

Experimentation to evaluate the effect of priming treatments under soil tubes

To assess the effectiveness of the priming treatments under field conditions, we simulated soil tube method (Tube diameter 2.5 cm & Tube length 30 cm); in which soil with 100% (Matric Potential–35 kPa) field capacity was kept as control in comparison with 75% (Matric Potential–170 kPa) and 60% (Matric Potential–430 kPa) of field capacity as moisture stress situation. The soil (vertisol) was collected from 0 to 20 cm depth and characterized with 62.8% clay, 36.6% silt and 0.6% sand with 0.78% soil organic carbon and cation exchange capacity of 61.7. The pH of the soil was around 8.0.

The field capacity of the soil sample was determined following the methodology described [38]. A 200 g sample of homogenized, oven-dried soil (W_d) was placed in a 1-liter perforated container lined with filter paper. The sample was saturated with water and left to equilibrate for one hour to ensure uniform absorption. It was then allowed to drain freely over a collection vessel for 48 h to eliminate gravitational water. After draining, the wet soil was weighed (W_w=254 g). The field capacity was calculated to be 27% using the following equation.

$$FC (\%) = [(W_w - W_d) / W_d] \times 100$$

After that the soil with 75% and 60% FC were prepared accordingly as per the Eqs. 1 and 2, and experimental setup was done as represented in supplementary Fig. 2. The shoot and root length were recorded with various priming treatments along with untreated control (Fig. 5).

$$75\% \text{ FC : Required } (27/100) \times 75 = 202.5 \text{ g H}_2\text{O kg}^{-1} \text{ of soil} \quad (1)$$

$$60\% \text{ FC : Required } (27/100) \times 60 = 162 \text{ g H}_2\text{O kg}^{-1} \text{ of soil} \quad (2)$$

Statistical analysis

The data recorded in three replications for each parameter was used for further statistical analysis (one way and two way-ANOVA) with Randomized Block Design. Mean separation of various quantitative parameters was performed based on the Tukey's Honestly Significant Difference test with $\alpha=0.05$ level of significance in SAS Statistical programming software version 9.4 (SAS Institute Inc. 2016. SAS® 9.4 Language Reference: Concepts, Sixth Edition. Cary, NC: SAS Institute Inc). The figures

were prepared using the graph builder function of SAS software.

Results

Characterization of chitosan nanoparticles (CNPs)

Hydrodynamic size, zeta potential and polydispersity

The hydrodynamic size and zeta potential of the synthesized chitosan nanoparticles (CNPs) were determined using dynamic light scattering (DLS) with a Malvern Zetasizer Nano ZS90 at 25 °C. The CNPs exhibited an average hydrodynamic size of 128.9 nm with a polydispersity index (PDI) of 0.410 and a zeta potential of -18.6 mV (Fig. 1).

Transmission Electron Microscopy (TEM)

Transmission electron microscopy (Joel JEM 2100 Plus) micrographs of chitosan nanoparticles (CNPs) revealed that the particles were predominantly spherical to oval (ellipsoidal) in shape. Although some size variability was observed, the particles generally exhibited smooth, rounded contours without sharp or angular features. The background appeared relatively clean with sparsely distributed particles, indicating minimal aggregation which falls within the nanometre range. The nanoparticles displayed uniform contrast, suggesting a relatively homogeneous internal structure, which is characteristic of polymer-based nanoparticles. The particle size distribution of the synthesized chitosan nanoparticles ranged between 27.9 and 143.0 nm. (Fig. 2).

FTIR (Fourier Transform Infrared Spectroscopy)

The FTIR (Diamond alpha ATR, Bruker optics, Germany GmbH) spectrum confirms the chemical integrity and functional groups of the chitosan nanoparticles. A broad, intense absorption band at 3362 cm^{-1} is attributed to the overlapping O-H and N-H stretching vibrations, indicating extensive intermolecular hydrogen bonding. The peak at 2952 cm^{-1} corresponds to the aliphatic C-H

stretching of the Alkane group. The presence of residual acetyl groups is confirmed by the Amide I band (C=O stretching) at 1653 cm^{-1} , while the Amide II band appearing at 1600 cm^{-1} is associated with the N-H bending of the primary amine groups. In Chitosan nanoparticles, particularly those cross-linked with agents like TPP (Tri-polyphosphate), the intensity and position of this peak can shift, reflecting the interaction between the amino groups and the cross-linker. The peak at 1410 cm^{-1} and 1037 cm^{-1} represents the symmetric C-H bending (deformation) of the methylene groups and C-N stretching of amine groups respectively. These characteristic peaks collectively verify the successfully maintained chitosan structure following nanoparticle formulation (Fig. 3, supplementary Table 1).

UV Spectrophotometer

UV spectrochemistry characterization of chitosan nanoparticles was also performed. The absorption peak for chitosan nanoparticles was obtained at 222 nm (supplementary Fig. 3).

Efficacy of Polyethylene glycol (PEG) in inducing drought stress

Analysis of the data across priming treatments to isolate the effect of stress severity confirmed a significant dose-dependent negative impact of PEG on almost all parameters (Table 1). Increasing PEG concentration from 1% to 5% resulted in progressive and significant reductions in shoot length, root length, germination percentage, and both seed vigour indices. Conversely, the osmotic stress triggered a significant acclimatory biochemical response, manifesting as an increase in total protein content, Catalase, Super oxide dismutase and peroxidase activity in the 1% and 5% PEG treatments compared to the non-stressed control. This confirms a successful induction of drought stress and the activation of protective metabolic pathways in the seeds.

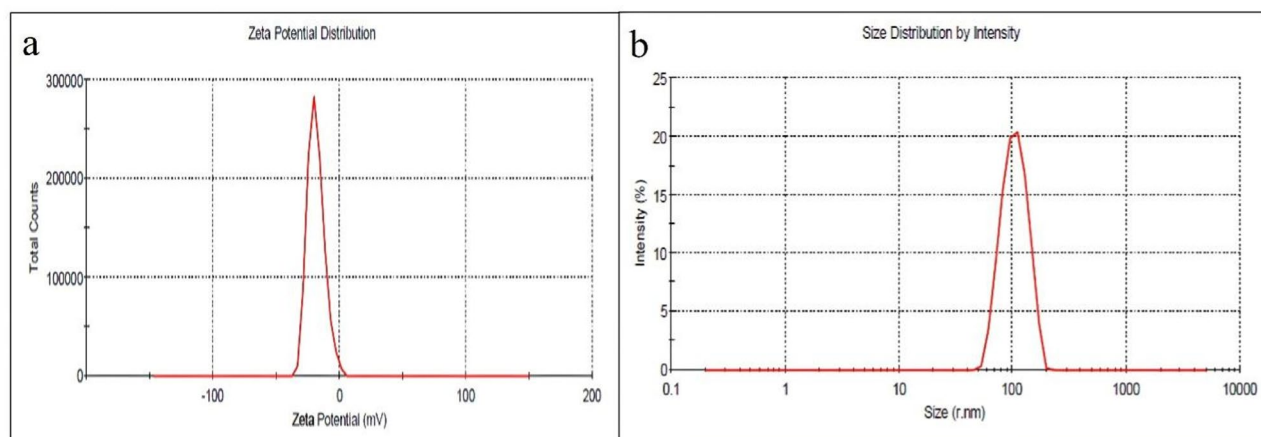


Fig. 1 Zeta potential (a) and particle size analysis of CNPs (b)

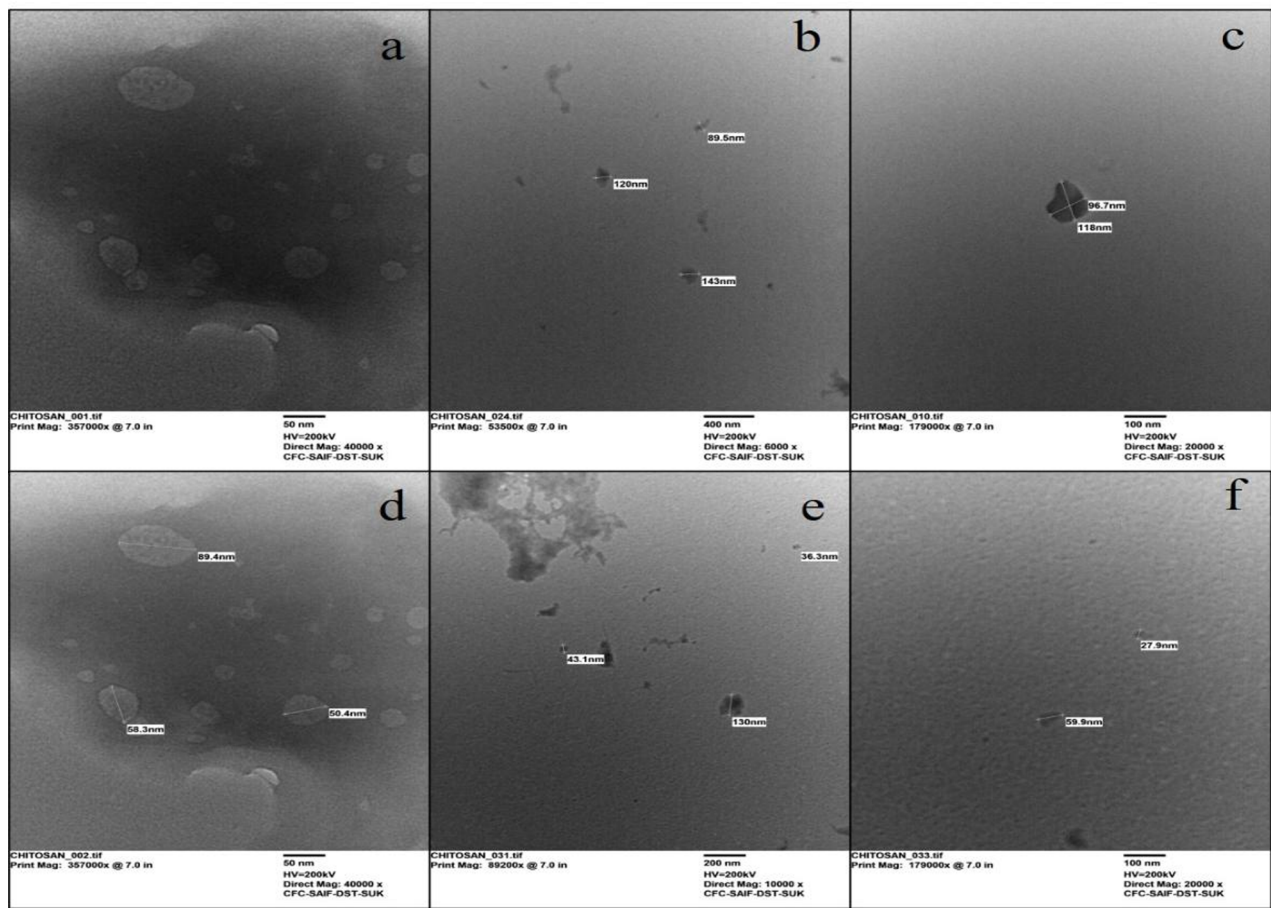


Fig. 2 TEM images of Chitosan nanoparticles (CNPs); **a-f**: representation of various size of nanoparticles

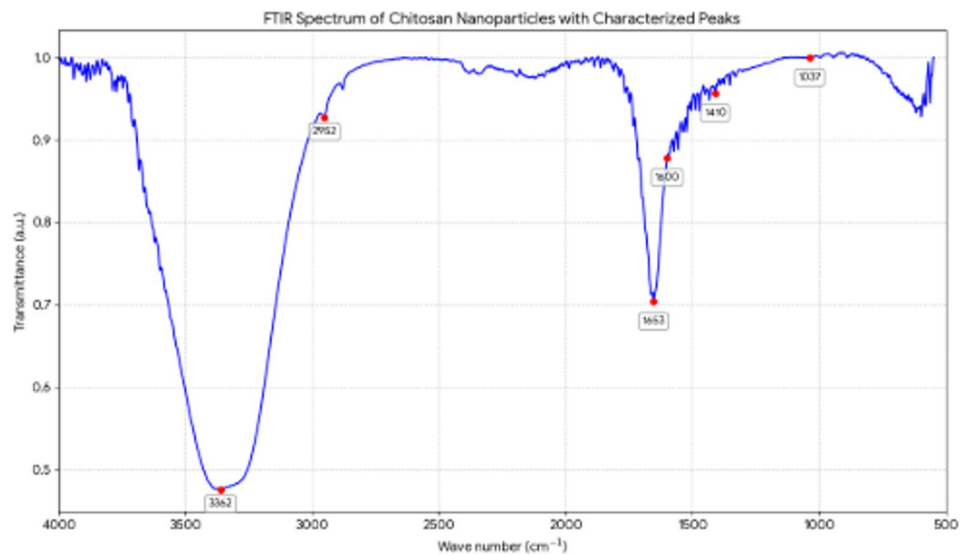


Fig. 3 FTIR characterization of synthesized chitosan nanoparticles

Table 1 Efficacy of PEG in inducing osmotic stress and the impact on overall seed vigour related traits

Treatment	Shoot Length (cm)	Root Length (cm)	Seedling Dry Weight (mg)	Germination (%)	SV-I	SV-II	Total Protein (mg/g)	Peroxidase (unit activity ml ⁻¹ min ⁻¹)	Catalase (unit activity ml ⁻¹ protein min ⁻¹)	Superoxide Dismutase (unit activity ml ⁻¹)
No Stress	12.96 ^a	11.18 ^a	832.22 ^a	95.33 ^a	2314.68 ^a	79602.22 ^a	1.55 ^b	0.030 ^b	0.031 ^a	1.56 ^b
1% PEG	11.73 ^b	10.24 ^{ab}	783.89 ^a	93.78 ^b	2069.90 ^b	73701.11 ^{ab}	1.87 ^a	0.033 ^a	0.025 ^a	1.71 ^a
5% PEG	10.59 ^c	9.36 ^b	740 ^a	92.0 ^c	1841.58 ^c	68292.22 ^b	1.91 ^a	0.034 ^a	0.04 ^a	1.63 ^{ab}

*Different letters within a particular column indicate significant difference ($p < 0.05$)

Table 2 Significance of experimental factors on seed vigour parameters

Sources of variation	df	Shoot Length (cm)	Root Length (cm)	Seedling Dry Weight (mg)	Germination (%)	SV-I	SV-II	Total Protein (mg/g)	Peroxidase (unit activity ml ⁻¹ min ⁻¹)	Catalase (unit activity ml ⁻¹ protein min ⁻¹)	Superoxide Dismutase (unit activity ml ⁻¹)
Priming treatment	2	**	**	*	**	**	**	**	**	**	**
Conditions	2	**	*	ns	**	**	*	*	**	ns	*
Priming treatment x Conditions	4	ns	ns	ns	ns	*	ns	ns	**	ns	**

df Degrees of freedom, SV-I Seed Vigour Index-I, SV-II Seed Vigour Index-II

* Significant at 5% level of significance ($p < 0.05$), ** Significant at 1% level of significance ($p < 0.01$)

Table 3 Efficacy of seed priming on seed germination, seedling growth and antioxidant enzyme activity

Treatment	Shoot Length (cm)	Root Length (cm)	Seedling Dry Weight (mg)	Germination (%)	SV I	SV II	Total Protein (mg/g)	Peroxidase (unit activity ml ⁻¹ min ⁻¹)	Catalase (unit activity ml ⁻¹ protein min ⁻¹)	Superoxide Dismutase (unit activity ml ⁻¹)
Untreated (control)	10.29 ^c	8.08 ^c	718.89 ^b	90.22 ^c	1657.78 ^c	64897.78 ^b	1.01 ^c	0.029 ^b	0.01 ^b	1.53 ^b
Hydropriming	11.94 ^b	10.01 ^b	766.11 ^b	92.89 ^b	2040.93 ^b	71247.78 ^b	1.86 ^b	0.032 ^b	0.03 ^{ab}	1.61 ^b
200 ppm CNPs	13.04 ^a	12.70 ^a	871.11 ^a	98.0 ^a	2527.44 ^a	85450.0 ^a	2.47 ^a	0.036 ^a	0.05 ^a	1.76 ^a

SV-I Seed Vigour Index-I, SV-II Seed Vigour Index-II

*Different letters within a particular column indicate significant difference ($p < 0.05$)

Impact of experimental factors on seed vigour parameters

Analysis of variance revealed that priming treatment had a significant effect on all measured seed vigour parameters ($p < 0.05$ or $p < 0.01$; Table 2). Specifically, it had a highly significant influence ($p < 0.01$) on shoot length, root length, germination, seed vigour index -I (SVI-I), seed vigour index-II (SV-II), total protein (mg/g), catalase, superoxide dismutase and peroxidase activity. Similarly, the stress conditions under which the seeds were tested also impacted the vigour parameters. The effect was highly significant ($p < 0.01$) on shoot length and germination, and significant ($p < 0.05$) for root length, SV-I, SV-II, total protein, and peroxidase activity except for seedling dry weight. The interaction between priming treatment and conditions was significant for SVI ($p < 0.05$), superoxide dismutase ($p < 0.01$) and peroxidase activity ($p < 0.01$). In contrast a non-significant interaction effect between priming treatment and conditions was observed for shoot length, root length, seedling dry weight, and germination, indicating that the main effect of priming was consistent across the stress conditions.

Effect of seed priming on seed germination, seedling growth and antioxidant enzyme activity

The effects of seed priming treatments on germination percentage, seedling vigour and enzymatic activity were investigated in the present study. All measured parameters differed significantly among the treatments ($p < 0.05$), with 200 ppm chitosan nanoparticle (CNPs) priming consistently outperforming both hydropriming and the untreated control (Table 3).

Germination percentage was significantly enhanced by chitosan nanoparticle (CNPs) priming (98.0%), compared to hydropriming (92.89%) and the untreated control (90.22%). Correspondingly, both seed vigour index-I (SV-I) and seed vigour index-II (SV-II) was significantly superior in chitosan nanoparticle (CNPs) primed seeds compared to hydropriming (SV-I: 2040.93; SV-II: 71,247.78) and the untreated control (SV-I: 1657.78; SV-II: 64,897.78). chitosan nanoparticle (CNPs) seedlings also exhibited the greatest shoot length (13.04 cm), root length (12.70 cm) and seedling dry weight (871.11 mg).

Total protein content (mg/g) was significantly highest in chitosan nanoparticle (CNPs) primed seedlings (2.47), followed by hydro primed (1.86) and untreated seedlings

(1.01). Antioxidant enzyme activities such as, peroxidase, catalase, and superoxide dismutase (SOD) were also significantly elevated under CNP priming. Peroxidase activity showed its highest activity in chitosan nanoparticle (CNPs) priming (0.036), significantly higher than in hydro priming (0.032) and untreated control (0.029). Catalase activity and SOD activity also followed the similar trend of peroxidase activity (Table 3) resulting significantly higher activity in chitosan nanoparticle priming.

Interaction effect of priming treatments under various PEG induced stress conditions

The effects of different priming treatments (Hydropriming and 200 ppm CNPs) on seedling growth and biochemical parameters under varying levels of PEG-induced stress (No stress, 1%, and 5%) were assessed in the present study through two factor (priming treatment and stress conditions) analysis. The data revealed the efficacy of 200 ppm CNPs significantly in maintaining growth and stress tolerance compared to hydropriming and untreated controls. The interaction effect of 200 ppm CNPs treatment and no stress condition (200 ppm CNPs x no stress) recorded the highest values across all seedling parameters.

Seedling growth traits

The efficacy of seed priming with chitosan nanoparticles (CNPs) in mitigating polyethylene glycol (PEG)-induced drought stress was evaluated by measuring a range of seed vigour-related parameters and biochemical parameters. Seed priming treatments significantly influenced seedling growth under both non-stress control condition as well as under PEG-induced stress conditions (Fig. 4). Under non-stress conditions, priming with 200 ppm CNPs resulted in the most robust growth, producing seedlings with significantly greater root length (14.60 cm), shoot length (14.97 cm), and dry weight (940 mg) compared to the hydropriming and untreated control groups (Table 4).

The imposition of drought stress (1% and 5% PEG) led to a marked reduction in all growth parameters across all treatments. However, CNPs priming consistently and effectively ameliorated this stress-induced growth inhibition. At the highest stress level (5% PEG), CNPs primed seedlings maintained superior root length (10.80 cm), shoot length (11.27 cm), and dry weight (823.33 mg), values which were significantly higher than those of the stressed untreated control. Although, seedling dry weight was statistically on par under all the priming treatments. Hydropriming showed an intermediate effect, generally outperforming the untreated control but failing to match the efficacy of CNPs treatment.

Seed vigour and germination potential

While PEG stress generally reduced seed vigour metrics, seeds treated with 200 ppm CNPs mitigated these effects effectively. For instance, the interaction between high stress (5% PEG), and the 200 ppm CNPs treatment (5% PEG x 200 ppm CNPs) maintained a germination percentage of 96.33%, whereas untreated seeds reduced to 88.67% under 5% PEG induced drought stress. The interaction was also statistically significant for where SV I peaked at 2126.33 for CNPs priming, compared to only 1778.37 for untreated seeds. The interaction between priming treatment and stress conditions was non-significant although, Seed vigour index II (SVI II) was also superior under the CNPs primed seeds under no stress, compared to other groups (Table 4).

In the absence of stress (no stress conditions), the 200 ppm CNPs treatment yielded near-complete germination (99.67%) and highest vigour (SV-I: 2947.33; SV-II: 93700). Hydro primed seeds recorded with second highest germination% and seed vigour index (I and II) compared to untreated control (Table 4). Drought stress caused a significant decline in vigour and germination across all groups. Nevertheless, seeds primed with CNPs exhibited remarkable resilience. This demonstrates a pronounced protective effect of CNPs on seed metabolic processes essential for vigour, even under severe water deficit.

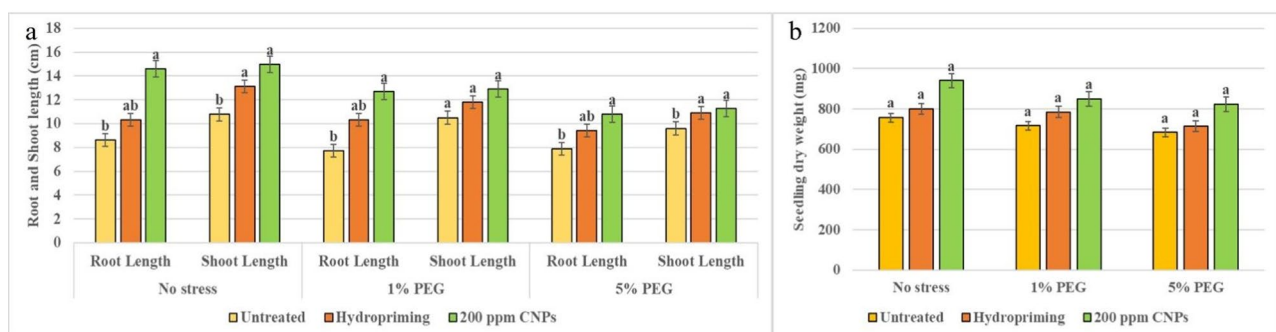


Fig. 4 Mean effect of priming treatments on seedling traits under PEG induced stress and control conditions; *Different letters within a particular stress condition for each trait indicate significant difference ($p < 0.05$); CNPs: Chitosan nanoparticles; **a**: Root and shoot length (cm); **b**: seedling dry weight(mg)

Table 4 Two-way interaction effect of priming treatments on seedling traits under PEG induced stress and control conditions

Priming Treatment	Conditions	Shoot Length (cm)	Root Length (cm)	Seedling Weight (mg)	Germination (%)	SV-I	SV-II	Total Protein (mg/g)	SOD (unit activity ml ⁻¹)	CAT (unit activity ml ⁻¹ protein min ⁻¹)	POX (unit activity ml ⁻¹ min ⁻¹)
Untreated (control)	No stress	10.77 ^{cd}	8.63 ^{cd}	756.67 ^{ab}	91.67 ^{ef}	1778.37 ^{de}	69380.0 ^{bc}	0.776 ^f	1.300 ^c	0.0153 ^a	0.0303 ^d
	1% PEG	10.50 ^d	7.73 ^d	716.67 ^{ab}	90.33 ^{fg}	1646.17 ^e	64713.33 ^{bc}	1.130 ^{ef}	1.650 ^{ab}	0.0103 ^a	0.0323 ^{bcd}
	5% PEG	9.60 ^d	7.86 ^d	683.33 ^b	88.67 ^g	1548.80 ^e	60600.0 ^c	1.136 ^{ef}	1.666 ^{ab}	0.0250 ^a	0.0270 ^d
Hydropriming	No stress	13.13 ^{ab}	10.30 ^{bcd}	800.0 ^{ab}	94.67 ^{cd}	2218.33 ^{bc}	75726.67 ^{abc}	1.586 ^{de}	1.704 ^{ab}	0.0330 ^a	0.0300 ^d
	1% PEG	11.80 ^{bcd}	10.30 ^{bcd}	785.0 ^{ab}	93.0 ^{de}	2054.87 ^{cd}	73083.33 ^{abc}	2.024 ^{bcd}	1.673 ^{ab}	0.0256 ^a	0.0303 ^d
	5% PEG	10.90 ^{cd}	9.43 ^{cd}	713.33 ^{ab}	91.0 ^{efg}	1849.60 ^{cde}	64933.33 ^{bc}	1.940 ^{cd}	1.453 ^{bc}	0.0356 ^a	0.0363 ^{abc}
200 ppm CNPs	No stress	14.97 ^a	14.60 ^a	940.0 ^a	99.67 ^a	2947.33 ^a	93700.0 ^a	2.296 ^{abc}	1.690 ^{ab}	0.0507 ^a	0.0320 ^{cd}
	1% PEG	12.90 ^{abc}	12.70 ^{ab}	850.0 ^{ab}	98.0 ^{ab}	2508.67 ^b	83306.67 ^{ab}	2.444 ^{ab}	1.818 ^a	0.0403 ^a	0.0380 ^{ab}
	5% PEG	12.27 ^{bcd}	10.80 ^{bc}	823.33 ^{ab}	96.33 ^{bc}	2126.33 ^{bcd}	79343.33 ^{abc}	2.653 ^a	1.761 ^{ab}	0.0607 ^a	0.0400 ^a

SV-I Seed Vigour Index-I, SV-II Seed Vigour Index-II, SOD Super Oxide Dismutase, CAT Catalase, POX Peroxidase

*Different letters within a particular column indicate significant difference ($p < 0.05$)

Biochemical responses

The biochemical profile of the seeds was significantly altered by both priming and stress treatments. Total protein content (mg/g) increased in response to PEG-induced stress in all treatments, with the highest concentrations consistently observed in the CNPs primed seeds across all conditions (No stress: 2.297 mg/g; 5% PEG: 2.653 mg/g). Peroxidase enzyme activity, a key marker of oxidative stress response, also showed a treatment-dependent pattern. While stress levels induced variability, the 200 ppm CNPs treatment consistently registered the highest peroxidase activities under stress conditions (Table 4), suggesting a priming-induced enhancement of the antioxidant defence mechanism. In case of catalase activity, under no stress condition 200 ppm CNPs primed seeds showed significantly highest activity compared to untreated seeds though, CNPs primed catalase activity was at par with hydro primed seed. On the other hand, catalase activity was statistically at par irrespective of seed priming treatments under 1% and 5% stress conditions. Superoxide dismutase activity showed a similar trend as of catalase under no stress conditions, whereas under 5% PEG induced stress 200 ppm CNPs primed seed showed significantly higher SOD activity; on contrary under 1% stress condition SOD activity was on par irrespective of priming treatments.

Overall, the 200 ppm CNPs treatment proved to be the most effective intervention, maintaining superior seedling growth, vigour, and antioxidant enzyme activity even under simulated drought stress conditions (Table 4).

Effect of priming treatments under soil tubes for shoot and root length

The results clearly showed priming treatments, particularly with 200 ppm chitosan nanoparticles (CNPs), significantly improve both root and shoot growth of the seedlings, especially under optimal soil moisture conditions (100% field capacity). The untreated seedlings showed the lowest root and shoot length across all field capacities (FC), with the best growth occurring at 75% field capacity (FC). In contrast, hydropriming and 200 ppm CNPs priming treatments consistently resulted longer roots and shoots under depleted soil moisture conditions also. The positive effect of priming treatments is most pronounced at the highest soil moisture level (100% FC). Seeds treated with 200 ppm CNPs, the root and shoot lengths under 60% and 75% FC were significantly greater than those of untreated seeds grown in 100% FC (Figs. 5 and 6). This indicated that CNPs priming effectively enhances plant growth by promoting better root development, which likely improves water and nutrient uptake.

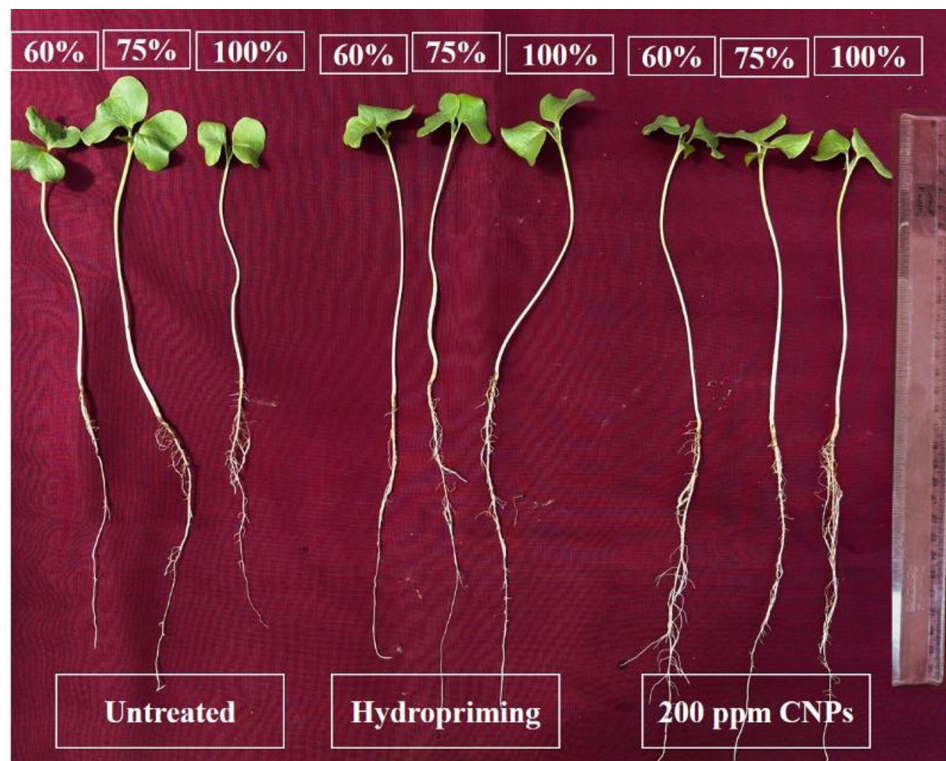


Fig. 5 Response of different priming treatment under soil tube at various field capacity (60% FC, 75% FC and 100% FC); FC: field capacity; CNPs: chitosan nanoparticles

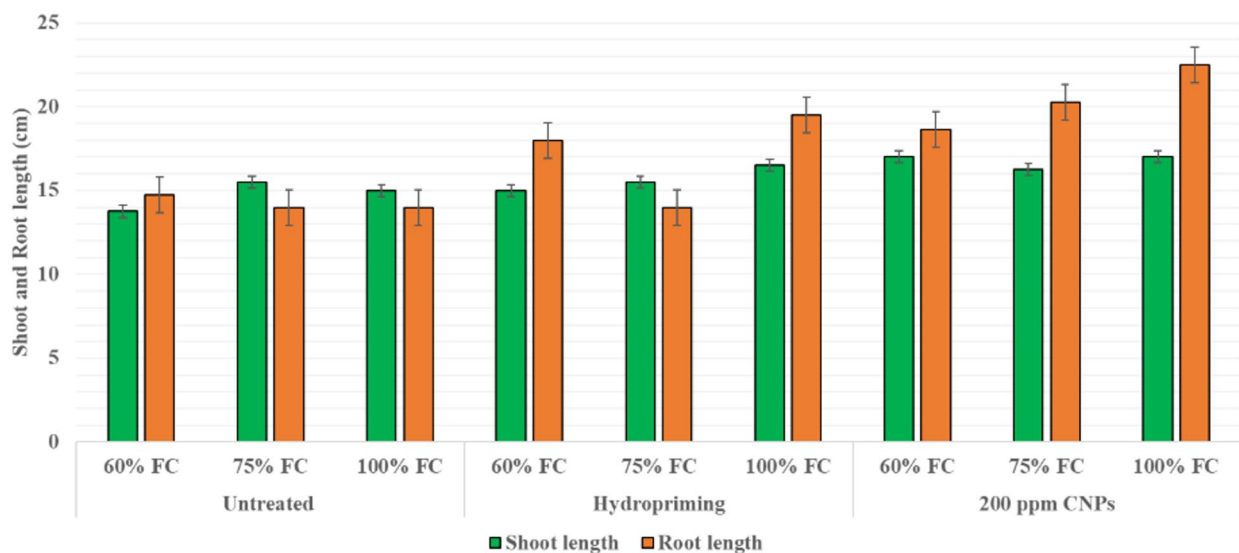


Fig. 6 Effect of priming treatments under soil tube conditions on root and shoot growth

Discussion

Assessment of chitosan nanoparticles

The present study demonstrated seed priming, particularly with chitosan nanoparticles (CNPs), is a highly effective strategy for enhancing drought tolerance in cotton. In our study, the CNPs exhibited an average hydrodynamic size of 128.9 nm with a polydispersity index

(PDI) of 0.410 and a zeta potential of -18.6 mV. These characteristics were compared to similar nanoparticles reported in the literature [39] synthesized CNPs with a larger hydrodynamic size of 373.5 nm, a PDI of 0.471, and a zeta potential of +45.3 mV. Similarly, previous studies [29] reported methyl jasmonate-chitosan nanoparticles (MJ-CNPs) with a hydrodynamic size of 296.1 nm, a PDI

of 0.4, and a zeta potential of +41.7 mV. Zeta potential (surface charge), which is an important parameter which influences the stability of nanoparticles [29]. A zeta potential below -30 mV and above +30 mV is an essential requirement to maintain a physically stable nanoparticle suspension [40].

The particle size distribution of the synthesized chitosan nanoparticles ranged between 27.9 and 143.0 nm; which falls within the nanometre range, confirming their nanoscale dimensions (Fig. 2). This low level of agglomeration, along with the consistent morphology and size distribution, suggests that the CNPs are likely homogeneous and monodispersed, making them suitable for applications. The discrepancy between TEM and DLS measurements is attributed to their differing principles; TEM provides the dry core size of individual nanoparticles, whereas DLS measures the hydrodynamic diameter in suspension, including the core, surface coatings, and surrounding solvent layers, typically resulting in larger size values [41].

Impact of PEG on inducing drought stress

The current investigation showed that drought stress imposed through polyethylene glycol substantially impaired seed germination and early seedling growth, which is consistent with earlier reports highlighting that osmotic stress limits seed water uptake, delays metabolic activation, and reduces radicle emergence [42, 43]. The negative impact of increasing PEG concentration on seedling growth and vigour both in paper towels (Table 1) validates the robustness of stress induction protocol. Moreover, PEG-induced stress can increase oxidative damage and impair the functioning of mitochondria and chloroplasts, causing a shift in cellular priorities from osmolyte accumulation to basic survival and repair processes. The observed stress-induced increase in total protein and peroxidase activity across treatments is a typical acclimation response due to chitosan nanoparticle priming. However, the fact that CNPs primed seeds reached the highest levels of these biomarkers underscores that priming does not merely trigger a generic stress response but amplifies it significantly, leading to a superior state of stress resilience. Polyethylene Glycol (PEG 6000) used in this study for inducing moisture stress previously used by earlier researchers to simulate drought condition during seed germination [6, 44].

Efficacy of seed priming treatments

Among the seed priming treatments, CNPs priming consistently outperformed the hydropriming and untreated control groups across seed vigour related parameters (seed vigour index I, seed vigour index-II, germination percentage, seedling dry weight) and biochemical parameters both under stress as well as normal germination

condition. While hydro-priming (with water) may only provide temporary hydration without biochemical activation. Chitosan and nano-chitosan not only regulate osmotic stress but also, act as biostimulants, promoting plant growth and enhancing resistance to biotic and abiotic stresses [45]. Under stress condition CNPs priming mitigate the inhibitory effects of osmotic stress on metabolic processes that maintained germination percentage of chitosan nanoparticle primed seeds compared to untreated control which may be due to increased reserve mobilization and energy production [39, 46]. This report also supported our findings on enhanced seedling growth and biomass accumulation in CNPs primed cotton seeds i.e. the increased vigour indices, both SV I, SV II and germination rate under severe osmotic stress (5% PEG). This is a key attribute for crop establishment in arid and semi-arid environments where soil moisture is limited and unpredictable.

Seed vigour parameters

The significant improvement in root and shoot length, along with dry weight was observed in CNPs primed seeds especially under stress conditions. An earlier finding by researchers suggested that beneficial effects of priming are most evident under stress conditions though advantages are also observed at optimal germination condition [47]. Increase in root length is critical since properly developed root system enables the plant to absorb water from deeper soil layers and helps in climate resilience [48, 49]. Our study on effect of seed priming in soil tubes simulated the field condition supporting the efficacy of CNPs for superior root development and drought mitigation in cotton.

Chitosan nanoparticle (CNPs) priming maintained the seed germination under stressful environment as observed in the present study (Table 3) compared to untreated seeds under stress; is important efficiency indicator of chitosan priming since germination is a transition stage from seed to seedlings and, higher the seed germination under stressful conditions, higher the ability of plants later to thrive and produce higher yields under adverse environments [50].

Biochemical parameters

The highest total protein content consistently recorded in CNPs primed seeds across all conditions revealed biochemical changes occurring enabling seed to withstand stress which was in parallel with a study in mung beans where an increase in protein content in CNPs primed seeds with an increase in stress also reported [27]. It has been revealed that plants often upregulate the synthesis of specific proteins under stress, including late embryogenesis abundant (LEA) proteins and osmo-protectants, which stabilize cellular structures and maintain osmotic

balance [51, 52]. The significantly higher protein content in CNPs treatments suggests a priming-induced “ready-state,” enabling a more efficient and robust protective response to impending water deficit.

The elevated peroxidase and superoxide dismutase activity observed in CNPs primed seeds under stress is another key factor supporting biochemical changes taking place in primed seeds. Drought stress inevitably leads to the accumulation of reactive oxygen species (ROS) that cause oxidative damage to lipids, proteins, and nucleic acids. The enhanced peroxidase activity of nano-primed seeds observed in our study indicates that CNPs priming pre-activates the antioxidant machinery, allowing the seeds to swiftly and effectively detoxify ROS upon stress imposition. The efficacy of seeds primed with nanoparticles in drought amelioration and enhanced activity of anti-oxidant enzymes including peroxidase on primed seeds was reported [26, 60]. The concurrent increase in both protective proteins and antioxidant enzyme activity points to a multi-faceted mechanism where CNPs bolster cellular defence systems [61]. Peroxidase is a crucial enzyme in the antioxidant defence system that scavenges reactive oxygen species (ROS). Previous researchers also established chitosan nanoparticle (CNPs) priming can alter gene expression, demonstrating the effects at the genetic and molecular level. Specifically, CNPs have been shown to modulate the expression of drought-responsive genes, including those involved in proline biosynthesis and key dehydration-responsive transcription factors such as DREB (Dehydration Responsive Element Binding) proteins. These proteins function as critical regulators by binding to DRE elements in the promoter regions of drought inducible genes. CNPs treatment enhances DREB1A expression, which in turn activates downstream genes such as RD29A, COR15 and LEA proteins [63].

Soil simulation study

The soil tube study demonstrated that seed priming with 200 ppm chitosan nanoparticles (CNPs) significantly resulted root elongation, across varying soil moisture regimes. Enhanced root development under water-deficit conditions (60–75% FC) aligns with findings that nanoparticle-based priming upregulates aquaporin expression and antioxidant defence mechanisms [53]. Hydropriming showed intermediate benefits, consistent with established earlier literatures on hydration mediated metabolic activation [54]. The superiority of chitosan nanoparticles over hydropriming at reduced field capacity (moisture stress) suggested chitosan nanoparticles confer stress memory, enabling sustained growth promotion under suboptimal moisture availability. Earlier researcher [55] also noted the role of chitosan and nano-chitosan as priming agents in mitigating abiotic stress.

Conclusion

In conclusion, seed priming with 200 ppm chitosan nanoparticles emerges as a superior strategy to enhance drought tolerance. It acts through a dual mechanism, one by promoting robust root and shoot development for improved water relations, and through pre-activating the plant's biochemical defence system, including the synthesis of protective proteins and antioxidant enzymes, to mitigate osmotic and oxidative damage. Although in future the whole metabolite profiling of chitosan nanoparticle primed seeds compared to untreated seeds may be undertaken to understand the mechanism of enhanced seed vigour performance of CNP treated seeds more clearly. These findings indicate chitosan nanoparticle (CNPs) priming as a highly promising, nano-enabled approach for improving cotton crop stand establishment and productivity in water-limited environments. In future additional biochemical markers such as lipid peroxidation (malondialdehyde, MDA content), glutathione reductase (GR) activity and ascorbate peroxidase (APX) activity may be taken up for better elucidating the mechanistic basis of chitosan nanoparticles in drought tolerance.

Abbreviations

CNPs	Chitosan Nanoparticles
TTP	Trisodium Polyphosphate
Bt	Bacillus thuringiensis
ROS	Reactive Oxygen Species
PEG	Polyethylene Glycol
FTIR	Fourier Transform Infrared Spectroscopy
TEM	Transmission Electron Microscopy
DLS	Dynamic Light Scattering
CAT	Catalase
POX	Peroxidase
SOD	Superoxide Dismutase
FC	Field Capacity
DREB proteins	Dehydration Responsive Element Binding proteins

Supplementary Information

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Supplementary Material 1.

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Plant guideline statement

Experimental research and field studies on plants (either cultivated or wild) was as per the relevant institutional guidelines and legislation of ICAR-CICR, Nagpur, Maharashtra, India. No leaves samples were collected.

Authors' contributions

VS, DP: Conceptualization; VS, PD, RK, AM: Conduct of experiments; DP: Data analysis; DP, PD: preparation of tables and figures; VS, PD, DP, RK, VNW: writing the first draft of the manuscript; All authors: editing and proof reading of the manuscript.

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Data availability

All the data are available in the manuscript.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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References

- Saleem MF, Sammar Raza MA, Ahmad S, Khan IH, Shahid AM. Understanding and mitigating the impacts of drought stress in cotton-a review. *Pak J Agric Sci*. 2016;53(3):609–23.
- Meshram JH, Singh SB, Raghavendra KP, Waghmare VN. Chapter 6 - Drought stress tolerance in cotton: progress and perspectives, Editor(s): Arun K. Shanker, Chitra Shanker, Anjali Anand, M. Maheswari, Climate Change and Crop Stress, Academic Press, 2022, Pages 135–169, ISBN 9780128160916. <https://doi.org/10.1016/B978-0-12-816091-6.00005-5>.
- Shani MY, Ditta A, Khan MKR, Rahimi M. Deciphering drought tolerance in cotton genotypes through integrated morpho-physiological and biochemical markers at flowering stage. *Sci Rep*. 2025;15:44123. <https://doi.org/10.1038/s41598-025-28237-6>.
- Tian Y, Wang Z, Ma P, Zhao Z, Xie Z. Drought priming may enhance the tolerance of cotton seedlings to subsequent drought stress. *Plant Mol Biol Rep*. 2024;42(4):625–38. <https://doi.org/10.1007/s11105-024-01437-6>.
- Memon S, Abro A, Jakhro MI, Farid A, Habib M, Ahmed M, Bhutto LA, Memon SA, Farooq M. Polyethylene glycol mediated osmotic stress impacts on growth and biochemical aspects of wheat under artificial osmotic stress condition. *J innov sci*. 2023;9(1):44–50. <https://doi.org/10.17582/journal/jis/2023/9.1.44.50>.
- Hossain MS, Li J, Wang C, Monshi FI, Tabassum R, Islam MA, Faruquee M, Muktadir MA, Mia MS, Islam AKMM, Hasan AK, Sikdar A, Feng B. Enhanced Antioxidant Activity and Secondary Metabolite Production in Tartary Buckwheat under Polyethylene Glycol (PEG)-Induced Drought Stress during Germination. *Agronomy*. 2024;14(3):619. <https://doi.org/10.3390/agronomy14030619>.
- Peršić V, Ament A, Antunović Dunić J, Drezner G, Cesar V. PEG-induced physiological drought for screening winter wheat genotypes sensitivity-integrated biochemical and chlorophyll a fluorescence analysis. *Front Plant Sci*. 2022;13:987702. <https://doi.org/10.3389/fpls.2022.987702>.
- Jincy M, Prasad V, Jeyakumar P, Senthil A, Manivannan N. Evaluation of green gram genotypes for drought tolerance by PEG (polyethylene glycol) induced drought stress at seedling stage. *Legume Res*. 2021; 44(6), 684–691. <https://doi.org/10.18805/LR-4149>.
- Iqbal A, Huiping G, Qiang D, et al. Differential responses of contrasting low phosphorus tolerant cotton genotypes under low phosphorus and drought stress. *BMC Plant Biol*. 2023;168. <https://doi.org/10.1186/s12870-023-04171-5>.
- Aswathi KPR, Sen A, Puthur JT. Comparative Study of Cis- and Trans-Priming Effect of PEG and BABA in Cowpea Seedlings on Exposure to PEG-Induced Osmotic Stress. *Seeds*. 2023;2(1):85–100. <https://doi.org/10.3390/seeds201007>.
- Dharshini AP, Rajendra Prasad VB, Vanitha K, Manivannan N. Effect of PEG Induced Drought Stress on Seed Germination and Seedling Growth of Green gram Genotypes. *Int. J. Environ. Clim. Chang*. 2021; 11(10), 79–90. <https://doi.org/10.9734/ijeccl/2021/v11i1030495>
- Reyes JAO, Casas DE, Gandia JL, Parducho MJL, Renovalles EM, Quillooy EP, Del-fin EF. Polyethylene glycol-induced drought stress screening of selected Philippine high-yielding sugarcane varieties. *J Agric Food Res*. 2023;14:100676. <https://doi.org/10.1016/j.jafr.2023.100676>.
- Saha D, Choyal P, Mishra UN, Dey P, Bose B, Gupta NK, Mehta BK, Kumar P, Pandey S, Chauhan J, Singhal RK. Drought stress responses and inducing tolerance by seed priming approach in plants. *Plant Stress*. 2022;4:100066. <https://doi.org/10.1016/j.stress.2022.100066>.
- Aswathi KPR, Kalaji HM, Puthur JT. Seed priming of plants aiding in drought stress tolerance and faster recovery: a review. *Plant Growth Regul*. 2022;97:235–53. <https://doi.org/10.1007/s10725-021-00755-z>.
- Paparella S, Araújo SS, Rossi G, Wijayasinghe M, Carbonera D, Balestrazzi A. Seed priming: state of the art and new perspectives. *Plant Cell Rep*. 2015;34(8):1281–93. <https://doi.org/10.1007/s00299-015-1784-y>.
- Shang Y, Hasan MK, Ahammed GJ, Li M, Yin H, Zhou J. Applications of nanotechnology in plant growth and crop protection: a review. *Molecules*. 2019;24(14):2558. <https://doi.org/10.3390/molecules24142558>.
- Mahdy AM, Sherif FK, Elkhatib EA, Fathi NO, Ahmed MH. Seed priming in nanoparticles of water treatment residual can increase the germination and growth of cucumber seedling under salinity stress. *J Plant Nutr*. 2020;43(12):1862–74. <https://doi.org/10.1080/01904167.2020.1750647>.
- Rai-Kalal P, Jajoo A. Priming with zinc oxide nanoparticles improve germination and photosynthetic performance in wheat. *Plant Physiol Biochem*. 2021;160:341–51. <https://doi.org/10.1016/j.plaphy.2021.01.032>.
- Shah T, Latif S, Saeed F, Ali I, Ullah S, Alsahli AA, Jan S, Ahmad P. Seed priming with titanium dioxide nanoparticles enhances seed vigor, leaf water status, and antioxidant enzyme activities in maize (*Zea mays* L.) under salinity stress. *J King Saud Univ Sci*. 2021;33(1):101207. <https://doi.org/10.1016/j.jksus.2020.10004>.
- Waqas Mazhar M, Ishtiaq M, Hussain I, Parveen A, Hayat Bhatti K, Azeem M, Thind S, Ajai M, Maqbool M, Sardar T, Muzammil K. Seed nano-priming with Zinc Oxide nanoparticles in rice mitigates drought and enhances agronomic profile. *PLoS ONE*. 2022;17(3):0264967. <https://doi.org/10.1371/journal.pone.0264967>.
- Mazhar MW, Ishtiaq M, Maqbool M, Ullah F, Sayed SR, Mahmoud EA. Seed priming with iron oxide nanoparticles improves yield and antioxidant status of garden pea (*Pisum sativum* L.) grown under drought stress. *S Afr J Bot*. 2023;162:577–87. <https://doi.org/10.1016/j.sajb.2023.09.047>.
- Donia DT, Carbone M. Seed Priming with Zinc Oxide Nanoparticles to Enhance Crop Tolerance to Environmental Stresses. *Int J Mol Sci*. 2023;24(24):17612. <https://doi.org/10.3390/ijms242417612>.
- Zhang H, Gao W, Wang N, Wang X, Ma X, Chen J, Tang Q, Zhang J. Nanopriming with carbon dots enhances cotton seed germination and salt tolerance by activating salt-induced ROS signaling to modulate Na⁺ homeostasis. *Plant Nano Biol*. 2025;100225. <https://doi.org/10.1016/j.plana.2025.100225>.
- An J, Hu P, Li F, Wu H, Shen Y, White JC, Tian X, Li Z, Giraldo JP. Emerging investigator series: molecular mechanisms of plant salinity stress tolerance improvement by seed priming with cerium oxide nanoparticles. *Environ Sci : Nano*. 2020;7:2214–28. <https://doi.org/10.1039/D0EN00387E>.
- Malerba M, Cerana R. Chitosan effects on plant systems. *Int J Mol Sci*. 2016;17(7):996. <https://doi.org/10.3390/ijms17070996>.
- Allam E, El-Darier S, Ghattass Z, Fakhry A, Elghobashy RM. Application of chitosan nanopriming on plant growth and secondary metabolites of *Pan-cratium maritimum* L. *BMC Plant Biol*. 2024;24(1):466. <https://doi.org/10.1186/s12870-024-05148-8>.
- Sen SK, Chouhan D, Das D, Ghosh R, Mandal P. Improvisation of salinity stress response in mung bean through solid matrix priming with normal and nano-sized chitosan. *Int J Biol Macromol*. 2020;145:108–23. <https://doi.org/10.1016/j.jbiomac.2019.12.170>.
- Moodi Z, Sahabi H, Feizi H, Shajari MA. The impact of chitosan nanoparticles on seedling and germination attributes and enzymatic activity of guar (*Cyamopsis tetragonoloba* L.) under osmotic stress. *Biocatal Agric Biotechnol*. 2024;61:103356. <https://doi.org/10.1016/j.jbcab.2024.103356>.
- Arya SS, Rookes JE, Cahill DM, Lenka SK. Chitosan nanoparticles and their combination with methyl jasmonate for the elicitation of phenolics and flavonoids in plant cell suspension cultures. *Int J Biol Macromol*. 2022;214:632–41. <https://doi.org/10.1016/j.jbiomac.2022.06.145>.
- Acharya P, Jayaprakasha GK, Crosby KM, Jifon JL, Patil BS. Nanoparticle-mediated seed priming improves germination, growth, yield, and quality of water-melons (*Citrullus lanatus*) at multi-locations in Texas. *Sci Rep*. 2020;10(1):5037. <https://doi.org/10.1038/s41598-020-61696-7>.

31. Paul D, Dutta S, Nidhi, Sapna Jha R, Biswas N, Bagri S, Kumar R. Heterosis breeding and hybrid seed production in cotton. In *Hybrid Seed Production for Boosting Crop Yields: Applications, Challenges and Opportunities* 2025; (pp. 225–46). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-96-0506-4_9
32. Abdul-Baki AA, Anderson JD. Vigor determination in soybean seed by multiple criteria. *Crop Sci.* 1973;13(6):630–3. [https://doi.org/10.1016/0003-2697\(76\)90527-3](https://doi.org/10.1016/0003-2697(76)90527-3). 0011183X001300060013x.
33. Bradford MM. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. *Anal Biochem.* 1976;72:248–54. [https://doi.org/10.1016/0003-2697\(76\)90527-3](https://doi.org/10.1016/0003-2697(76)90527-3).
34. Lurie S, Fallik E, Handros A, Shapira R. The possible involvement of peroxidase in resistance to *Botrytis cinerea* in heat treated tomato fruit. *Physiological Mol Plant Pathol.* 1997;50(3):141–9. <https://doi.org/10.1006/pmpp.1996.0074>.
35. Stewart RR, Bewley JD. Lipid peroxidation associated with accelerated aging of soybean axes. *Plant Physiol.* 1980;65:245–8. <https://doi.org/10.1104/pp.65.2.245>.
36. Beauchamp C, Fridovich I. Superoxide dismutase: Improved assay and an assay applicable to acrylamide gels. *Anal. Biochem.* 1971;44:276–87. [https://doi.org/10.1016/0003-2697\(71\)90370-8](https://doi.org/10.1016/0003-2697(71)90370-8).
37. Aebi H. Catalase in vitro. *Methods Enzymol.* 1984;105:121–6. [https://doi.org/10.1016/s0076-6879\(84\)05016-3](https://doi.org/10.1016/s0076-6879(84)05016-3).
38. Dane JH, Topp CG. Methods of soil analysis, Part 4: Physical methods. John Wiley & Sons; Soil Science Society of America, 2020; Inc. 667 south Segoe Road, Madison, WI 53711 – 1086, USA. SBN:9780891188414 Online ISBN: 9780891188933 <https://doi.org/10.2136/sssabookser5.4>
39. Soni AT, Rookes JE, Arya SS. Chitosan Nanoparticles as Seed Priming Agents to Alleviate Salinity Stress in Rice (*Oryza sativa* L). Seedlings Polysaccharides. 2023;4(2):129–1410. <https://doi.org/10.3390/polysaccharides4020010>.
40. Muthukrishnan S, Murugan I, Selvaraj M. Chitosan nanoparticles loaded with thiamine stimulate growth and enhances protection against wilt disease in Chickpea. *Carbohydr Polym.* 2019;212:169–77. <https://doi.org/10.1016/j.carbpol.2019.02.037>.
41. Adline FA, Nisa JD, Shanthi R, Sivakumar MR. Ionic gelation-based synthesis of chitosan nanoparticles from the waste shells of anomuran crab *Albunea symmista* (Linnaeus, 1758) exhibits potential applications. *Carbohydr Res.* 2025;109697. <https://doi.org/10.1016/j.carres.2025.109697>.
42. Queiroz M, Oliveira C, Steiner F, Zuffo A, Zoz T, Vendruscolo E, et al. Drought stresses on seed germination and early growth of maize and sorghum. *J Agric Sci.* 2019;11(2):310–8. <https://doi.org/10.5539/jas.v11n2p310>.
43. Mustamu NE, Tampubolon K, Alridiawirah Basyuni M, Al-Taey DKA, Al Janabi HJK, et al. Drought stress induced by polyethylene glycol (PEG) in local maize at the early seedling stage. *Heliyon.* 2023;9:e20209. 10.1016/j.heliyon.2023.e20209.
44. Eesha A, Sharma R, Chaudhary NS. Effect of PEG-6000 Osmoticum on Seed Germination and Seedling Growth of Lentil (*Lens culinaris* Medik.) Genotypes. *Indian J. Agric. Res.* 2024;58(3):456–461. <https://doi.org/10.18805/IJARE.A-6078>.
45. Farooq T, Nisa ZU, Hameed A, Ahmed T, Hameed A. Priming with copper-chitosan nanoparticles elicit tolerance against PEG-induced hyperosmotic stress and salinity in wheat. *BMC Chem.* 2022;16(1):23. <https://doi.org/10.1186/s13065-022-00813-1>.
46. Alhamadi MA, Gairola S, Alketbi A, Alketbi F, Hamdy R, Mosa KA, El-Keblawy A. Emerging progress in investigating the impact of green-synthesized nanoparticles on seed germination. *J Plant Res.* 2025;138:937–58.
47. Habibi N, Terada N, Pachakkil B, Sanada A, Kamata A, Koshio K. Effect of Priming Treatment on Improving Germination and Seedling Performance of Aged and Iron-Coated Rice Seeds Aiming for Direct Sowing. *Plants.* 2025;14(11):1683. <https://doi.org/10.3390/plants14111683>.
48. Kalra A, Goel S, Elias A. Prospects of Developing Drought Resilience in Crop Plants Through Strategic Modifications in Root System Architecture. *Authorea Preprints.* 2023. <https://doi.org/10.22541/essoar.167347082.25216669/v1>.
49. Behboudi F, Tahmasebi-Sarvestani Z, Kassaei MZ, Modarres-Sanavy SAM, Sorooshzadeh A, Mokhtassi-Bidgoli A. Evaluation of chitosan nanoparticles effects on yield and yield components of wheat (*Triticum aestivum* L.) under drought stress. *J Plant Nutr.* 2018;41(5):677–88. <https://doi.org/10.1080/01904167.2019.1617308>.
50. Tabatabai SMT, Goshasbi F, Bakhshi B. Evaluation of the effect of polyethylene glycol (PEG) on germination and morphological characteristics of bread wheat. *Cereal Res Commun.* 2022;41009–15. <https://doi.org/10.1007/s42976-022-00242-9>.
51. Hosseini SM, Hasanloo T, Mohammadi S. Physiological and molecular responses of plants to drought stress. *J Plant Res (Iranian J Biology).* 2021;34(1):1–20.
52. Kumar P, Paul D, Jhajhriya S, Kumar R, Dutta S, Siwach P, Das S. Understanding heat-shock proteins' abundance and pivotal function under multiple abiotic stresses. *J Plant Biochem Biotechnol.* 2024;33(4):492–513. <https://doi.org/10.1007/s13562-024-00932-x>.
53. Roychoudhury A, Chatterjee U, Bhattacharya A, Dutta S. Using Chitosan Nanoparticles in the Alleviation of Drought Stress in Plants. *Chitosan Nanoparticles for Plant Resilience Under Abiotic and Biotic Stressors*. Singapore: Springer Nature Singapore; 2026. pp. 109–46.
54. Farooq M, Hussain M, Siddique KH. Drought stress in wheat during flowering and grain-filling periods. *CRC Crit Rev Plant Sci.* 2014;33(4):331–49.
55. Sen SK, Das D. A sustainable approach in agricultural chemistry towards alleviation of plant stress through chitosan and nano-chitosan priming. *Discov Chem.* 2024;1:44. <https://doi.org/10.1007/s44371-024-00046-2>.
56. Wu S, Hu C, Tan Q, Li L, Shi K, Zheng Y, Sun X. Drought stress tolerance mediated by zinc-induced antioxidative defense and osmotic adjustment in cotton (*Gossypium hirsutum*). *Acta Physiol Plant.* 2015;37(8):167.
57. Bai Y, Xiao S, Zhang Z, Zhang Y, Sun H, Zhang K, Wang X, Bai Z, Li C, Liu L. Melatonin improves the germination rate of cotton seeds under drought stress by opening pores in the seed coat. *PeerJ.* 2020;8:e9450.
58. Jaafar KS, Mohammed MA, Mohammed SM. Screening for drought tolerance in cotton (*Gossypium hirsutum* L.) using in vitro technique. *J Dryland Agric.* 2021;7(4):52–9.
59. Moodi Z, Sahabi H, Feizi H, Shajari MA. The impact of chitosan nanoparticles on seedling and germination attributes and enzymatic activity of guar (*Cyamopsis tetragonoloba* L.) under osmotic stress. *Biocatal Agric Biotechnol.* 2024;61:103356.
60. Shakoor A, Saleem MF, Ul Haq MZ, Farooq MS, Shahid M. Chitosan Mitigates Heat Stress in BT Cotton by Activating Antioxidant Defense System and Improving Water Relations. *Int J Plant Prod.* 2026;20(1):6.
61. Zeng D, Luo X. Physiological effects of chitosan coating on wheat growth and activities of protective enzyme with drought tolerance. *Open J Soil Sci.* 2012;2(03):282.
62. Othman SH, Ronzi NDA, Shapi'i RA, Dun M, Ariffin SH, Mohammed MAP. Biodegradability of starch nanocomposite films containing different concentrations of chitosan nanoparticles in compost and planting soils. *Coatings.* 2023;13(4):777.
63. Choudhary Y, Rani N, Boora M, Devi S, Kumar A, Kumari S, Grewal S. Chitosan Nanoparticles and Their Conjugates in Drought Stress Management: A Cutting-Edge Strategy for Crop Resilience. *Plant Nano Biology.* 2026;15:100251.

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